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Predicting Voluntary Employee Attrition with Explainable Gradient Boosting: Evidence from Indian IT Services

R. Menon, S. Iyer, A. Demo · 2025 · *Journal of Business Research*

TL;DR — Using HR records of 18,420 employees from three mid-sized IT firms, we compare logistic regression, random forest and XGBoost.

0.89
AUC

18,420
employees

Secondary / archival data
research design

Machine-learning prediction
approach

RESEARCH PROBLEM

- However, most predictive attrition studies rely on the public IBM HR dataset, which is synthetic and small.
- Little is known about how well attrition models generalise to real Indian workforce data.

KEY FINDINGS

- XGBoost outperformed the other models with an AUC of 0.89, precision of 0.72 and recall of 0.81.
- SHAP analysis revealed that months since last promotion was the single most important driver, followed by commute distance above 25 kilometres and more than two manager changes in a year.
- The fairness audit found a difference of only 1.8 percentage points in false positive rates between men and women.

OBJECTIVES

- This study addresses both gaps.
- Our objective is to build an accurate and explainable early-warning model and to examine which organisational drivers matter most.

CONTRIBUTIONS

- HR managers can use the individual SHAP explanations to plan targeted interventions such as internal mobility offers and stable reporting lines.
- In addition, black-box predictions are rarely adopted by HR managers because they cannot explain why an employee is at risk.

METHODOLOGY

- Three models were trained: logistic regression, random forest and XGBoost.
- Voluntary attrition costs Indian IT services firms an estimated 1.5 to 2 times an employee's annual salary per exit.

LIMITATIONS

- The data come from only three firms in one industry, which limits generalisability.
- The model does not capture informal factors such as team climate or personal circumstances.

DATA & SAMPLE

- The dataset contains 18,420 employee records collected between 2019 and 2023 from three IT services firms in Bengaluru, Hyderabad and Pune.
- The voluntary attrition rate in the sample is 17.6 percent.

FUTURE RESEARCH

- Future research should test the model in other sectors, incorporate engagement survey text using natural language processing, and evaluate the causal effect of the recommended interventions through field experiments.

CITE (APA)

Menon, R., Iyer, S., & Demo, A. (2025). Predicting Voluntary Employee Attrition with Explainable Gradient Boosting: Evidence from Indian IT Services. *Journal of Business Research*, *172*(3), 114–129. https://doi.org/10.0000/demo.2025.001
Keywords: attrition, voluntary, model, explainable, indian, services, hr, firms · Methods: Logistic regression, Random forest, Gradient boosting, Explainable AI · Evidence support 1.00

AH

Generative AI Copilots and Analyst Productivity in Retail Banking: A Field Experiment

K. Rao, A. Demo, P. Sharma · 2026 · *Information Systems Research*

TL;DR — We report a randomised field experiment with 412 credit and product analysts at an Indian private-sector bank.

31%
task completion time by

412
analysts

Experimental
research design

Causal & experimental designs
approach

RESEARCH PROBLEM

- However, most evidence comes from software development and customer support, and little is known about regulated settings such as banking.
- A further challenge is over-reliance, where users accept incorrect AI output without verification.

KEY FINDINGS

- Quality scores improved by 9 percent on average.
- Junior analysts with less than three years of experience gained 43 percent in speed, compared with 18 percent for senior analysts.
- The copilot reduced average task completion time by 31 percent and increased output quality scores by 9 percent.

OBJECTIVES

- This paper examines whether a generative AI copilot improves analyst productivity and quality, and for whom the benefits are largest.

CONTRIBUTIONS

- A generative AI copilot substantially improved analyst productivity, but responsible deployment requires safeguards against over-reliance.
- Generative AI copilots act as a skill equaliser in analytical banking work.

METHODOLOGY

- The copilot used a large language model with retrieval over internal policy documents.
- We estimated treatment effects using difference-in-differences regression with analyst and week fixed effects.

LIMITATIONS

- The experiment was conducted in a single bank, and the twelve-week horizon cannot capture long-term skill erosion.
- Quality was assessed by internal reviewers who may share organisational biases.

DATA & SAMPLE

- The sample consists of 412 analysts, of whom 206 were in the treatment group.

FUTURE RESEARCH

- Future studies should examine long-run effects on analyst learning, replicate the design in public-sector banks, and test interface designs that reduce over-reliance.

CITE (APA)

Rao, K., Demo, A., & Sharma, P. (2026). Generative AI Copilots and Analyst Productivity in Retail Banking: A Field Experiment. *Information Systems Research*, *37*(2), 455–478. https://doi.org/10.0000/demo.2026.002
Keywords: ai, generative, analysts, copilot, reliance, percent, productivity, quality · Methods: Difference-in-differences, Randomised experiment, LLM / generative AI, Human evaluation · Evidence support 1.00

AH

Retrieval-Augmented Generation for Faithful Summarisation of Scientific Literature

L. Fernandes, M. Gupta · 2025 · *Information Processing & Management*

TL;DR — On a benchmark of 1,200 computer science and management papers, SecRAG improved factual consistency by 22 percent over direct prompting while maintaining ROUGE-L.

22%
improved factual consistency by

1,200
open-access papers

Design science / artefact
research design

Generative AI & NLP
approach

RESEARCH PROBLEM

- Automatic summarisation could help, but the problem of hallucination limits trust in LLM-generated summaries.

KEY FINDINGS

- The verification step alone removed 74 percent of injected false claims.
- In blind pairwise comparisons, 18 domain experts preferred SecRAG summaries in 68 percent of cases.
- Latency increased by 1.6 times because of the additional retrieval and verification calls.

OBJECTIVES

- This paper aims to design a summarisation method that is both concise and verifiably faithful.
- Large language models can summarise scientific papers fluently but frequently hallucinate numbers and claims.

CONTRIBUTIONS

- Section-aware retrieval and verification make LLM summaries of scientific papers substantially more faithful.
- Grounding each summary field in retrieved evidence trades a modest latency cost for a large gain in trustworthiness.

METHODOLOGY

- SecRAG first segments a paper into canonical sections such as introduction, methods and results.

LIMITATIONS

- The benchmark is limited to English papers, and figures and tables were not parsed.
- Automatic factuality metrics only approximate human judgement.

DATA & SAMPLE

- We built a benchmark of 1,200 open-access papers from arXiv and management journals, each paired with an author-written structured abstract.
- On a benchmark of 1,200 computer science and management papers, SecRAG improved factual consistency by 22 percent over direct prompting while maintaining ROUGE-L.

FUTURE RESEARCH

- Future work could extend SecRAG to multimodal content such as tables and charts, support multilingual papers, and personalise summaries to the reader's research profile.

CITE (APA)

Fernandes, L., & Gupta, M. (2025). Retrieval-Augmented Generation for Faithful Summarisation of Scientific Literature. *Information Processing & Management*, *62*(4), 103712. https://doi.org/10.0000/demo.2025.003
Keywords: papers, summaries, evidence, secrag, retrieval, summarisation, benchmark, verification · Methods: LLM / generative AI, Retrieval-augmented generation · Evidence support 1.00

Learning Analytics for Early Identification of At-Risk Postgraduate Management Students

N. Reddy, A. Demo · 2024 · *Computers & Education*

TL;DR — Mentor interventions triggered by the model reduced course failure by 19 percent in a pilot cohort.

0.77
F1

1,146
PGDM students

Quasi-experimental
research design

Machine-learning prediction
approach

RESEARCH PROBLEM

- However, mentors usually learn about struggling students only after end-term examinations.
- Learning analytics has been widely studied in undergraduate and online settings, but there is limited evidence from Indian postgraduate management education.

KEY FINDINGS

- The random forest model reached an F1 score of 0.77 by week five and 0.84 by week eight.
- The survival analysis showed that students who missed two consecutive assignment deadlines were 3.1 times more likely to disengage.
- In the pilot cohort, course failure fell by 19 percent compared with the previous cohort.

OBJECTIVES

- The purpose of this study is to examine how early at-risk students can be identified and whether model-triggered mentoring improves outcomes.
- This study builds an early-warning model for postgraduate management students using LMS activity, attendance and continuous assessment marks.

CONTRIBUTIONS

- Learning analytics can identify at-risk PGDM students early enough for mentors to act.
- Simple behavioural signals available early in the trimester are sufficient for useful predictions.

METHODOLOGY

- The model was retrained each week of the trimester to evaluate how early reliable predictions were possible.
- By week five of a trimester the model identified at-risk students with an F1 score of 0.77.

LIMITATIONS

- The pilot intervention was not randomised, so the reduction in failure may partly reflect cohort differences.
- The study used data from a single institution.

DATA & SAMPLE

- The dataset includes 1,146 PGDM students from four cohorts between 2021 and 2024.
- Data from 1,146 PGDM students across four cohorts were analysed with random forest and survival analysis.

FUTURE RESEARCH

- Future research should run randomised trials of mentoring interventions, include multiple institutions, and examine student perceptions of being flagged by algorithms.

CITE (APA)

Reddy, N., & Demo, A. (2024). Learning Analytics for Early Identification of At-Risk Postgraduate Management Students. *Computers & Education*, *208*, 105031. https://doi.org/10.0000/demo.2024.004
Keywords: students, early, failure, risk, learning, model, cohort, analytics · Methods: Random forest, Survival analysis · Evidence support 1.00

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Explainable Credit Scoring for Thin-File Borrowers Using Alternative Data

V. Krishnan, A. Demo, T. Bose · 2024 · *Finance Research Letters*

TL;DR — The alternative data model raised the Gini coefficient from 0.41 to 0.58 and could approve 14 percent more applicants at the same default rate.

0.41
Gini

52,300
loan applications

Secondary / archival data
research design

Machine-learning prediction
approach

RESEARCH PROBLEM

- Financial inclusion remains a policy priority in India.
- However, thin-file borrowers are often rejected because traditional scorecards cannot assess them.

KEY FINDINGS

- The LightGBM model with alternative data achieved a Gini coefficient of 0.58 compared with 0.41 for the bureau-only scorecard.
- The KS statistic improved from 0.31 to 0.44.
- At a constant default rate of 4 percent, the lender could approve 14 percent more applicants.

OBJECTIVES

- This paper investigates whether alternative data improves predictive power for thin-file borrowers and how decisions can be explained to applicants.
- This study evaluates whether alternative data such as utility payments and digital transaction patterns can improve credit scoring for thin-file borrowers while remaining explainable.

CONTRIBUTIONS

- Explainable machine learning with alternative data can responsibly improve credit access for thin-file borrowers.
- Alternative data can expand access to credit without increasing risk, provided that models are explainable and consent-based.

METHODOLOGY

- A LightGBM model was then trained on bureau variables plus alternative data features.
- The alternative data model raised the Gini coefficient from 0.41 to 0.58 and could approve 14 percent more applicants at the same default rate.

LIMITATIONS

- The study used data from one lender, and the out-of-time period did not include a severe economic downturn.
- Fairness across regions was not examined in depth.

DATA & SAMPLE

- The data comprise 52,300 loan applications from 2020 to 2023, of which 61 percent were thin-file applicants with fewer than two credit trade lines.
- Alternative data included utility bill payments, mobile recharge regularity and UPI transaction patterns, collected with applicant consent.

FUTURE RESEARCH

- Future work should assess model stability during economic stress, evaluate fairness across regions and income groups, and study borrower responses to counterfactual explanations.

CITE (APA)

Krishnan, V., Demo, A., & Bose, T. (2024). Explainable Credit Scoring for Thin-File Borrowers Using Alternative Data. *Finance Research Letters*, *61*, 104998. https://doi.org/10.0000/demo.2024.005
Keywords: data, alternative, credit, borrowers, thin file, explainable, fairness, model · Methods: Logistic regression, Gradient boosting, Explainable AI · Evidence support 1.00